Shih-Yu Lai

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*AI researcher with international experience in **sim-real learning and solvers** for high-performance, physical and embodied systems. My research spans **1. graphics**: simulation and animation production; **2. robotics**: manipulation, dynamics, and sensing; and **3. audio**: DSP and acoustics, focusing on physics-informed ML, neural PDE/ODE, reinforcement and robot learning, graph, generative/multimodal models.*

Summary

- Published **16 papers (Q1 journal:1 /conference: 11/preprints:4)** in top-tier AI/CS conferences during my *undergraduate/master studies*.
- Co-worked with **20+ nationalties** researchers from the *U.S. DoE, ETH Zurich, NUS, UBC, UCLA, UCL, TU Delft, UOsaka, Aalto, etc.*

Education

National Taiwan University

***M.S. in Computer Science** (Artificial intelligence Program, Admission Rate = **3**%)* *Expected Jun. 2027*

- Member @ Communications and Multimedia Lab | Advisor: Prof.Robin Bing-Yu Chen.
- *Collaborators:* Prof. Peter Yichen Chen (UBC), Dr. Tiantian Liu (Meshy AI), Dr. Ziqiu Zeng (NUS), Prof. Fan Shi (NUS), Prof. Yu-Lun "Alex" Liu (NYCU).

Publications (Selected: 9/16; [g: graphics, r: robotics, a: audio])

- **[g,r]** DiffPhD: A Unified Differentiable Solver for Projective Heterogeneous Materials in **Elastodynamics** with Contact-Rich GPU-Acceleration.
  Shih-Yu Lai, *Sung-Han Tien, Jui-I Huang, Yen-Chen Tseng, Yi-Ting Chiu, Siyuan Luo, Ziqiu Zeng, Fan Shi, Peter Yichen Chen, Tiantian Liu, Yu-Lun Liu, Bing-Yu Chen*, 2026, **ACM Transactions on Graphics (TOG), SIGGRAPH Asia 2026**, Paper / Project / Video / Code.
  #Real2Sim #Manipulation #Physical Simulation #Contact Dynamics #GPU Computing
- **[r]** MORPH-U: Multi-Objective Resilient **Motion Planning** for V2X-Enabled Autonomous Driving in High-Uncertainty Environments via **Simulation**.
  Shih-Yu Lai, 2026, **IEEE ITSC**, Paper.
  #Simulation #Motion Planning #Mobility #V2X #Optimization
- **[g,r]** Architectural CAD/CAM in Robotic Fabrication via Differentiable Energy Optimization and Few-Shot Spatiotemporal **Graph RL Policy**.
  Shih-Yu Lai, *Chia-Ching Yen, Yang-Ting Shen, Bing-Yu Chen*, 2025, **ACM SIGGRAPH Asia Technical Communications**, Paper.
  #Graph Reinforcement Learning #Robotic Manipulation #Differentiable Optimization #Hybrid Methods #Robotic Fabrication
- **[g,r] Graph Q-Learning** for Automatic Assembly in Design-to-Construction.
  Shih-Yu Lai, *Chia-Ching Yen, Yang-Ting Shen, Bing-Yu Chen*, 2025, **ACM SIGGRAPH Asia Posters**, Paper.
  #Reinforcement Learning #Graph Learning #Robotic Assembly #Manipulation #Learning Based Control
- **[g,r]** EUPHORIA: Efficient Universal Planning via Hybrid Optimization for Robust Industrial **Robotic Assembly**.
  Shih-Yu Lai, *Chia-Ching Yen, Yang-Ting Shen, Peter Yichen Chen, Yu-Lun Liu, Bing-Yu Chen*, 2026, **ICRA'27 Under Review**, Paper.
  #Robotic Assembly #Motion Planning #Hybrid Optimization #Industrial Robotics #Robust Planning
- **[a]** Estimation of **Room Impulse Responses** from Handclaps.
  Shih-Yu Lai, *Kyung Yun Lee, Nils Meyer-Kahlen, Eloi Moliner, Bing-Yu Chen, Vesa Välimäki*, 2027, **ICASSP, Under Review**, Paper.
  #Acoustic Measurements #Acoustic Signal Processing #Deep Learning #Inverse Problems #Reverberation
- **[a]** Latent-Mark: An **Audio Watermark** Robust to Neural Codec Compression.
  Yen-Shan Chen, ***Shih-Yu Lai***, *Ying-Jung Tsou, Yi-Cheng Lin, Bing-Yu Chen, Yun-Nung Chen, Hung-Yi Lee, Shang-Tse Chen*, 2026, **INTERSPEECH 2026**, Long Paper; Scores: 5,5,5,4.
  #Audio Watermarking #Neural Audio Codecs #Audio Signal Processing #Deep Learning #Robustness
- **[r]** UMEDA: Unified Multi-modal Efficient Data Fusion for Privacy-Preserving Graph Federated Learning via Spectral-Gated Attention and Diffusion-Based Operator Alignment.
  Shih-Yu Lai, *Hirozumi Yamaguchi, Shang-Tse Chen, Yu-Lun Liu, Bing-Yu Chen*, 2026, **ICLR'27 Under Review**, Paper.
  #Graph Learning #Multimodal Learning #Diffusion Models #Federated Learning #Representation Learning
- **[g]** MOONWALK: Mediating Operations with Intent-Evidence-Action Alignment Across Junior-Supervisor Review Workflows in **Animation/VFX Pre-Production**.
  Shih-Yu Lai, *Wen-Fan Wang, Sai Ling, Shaune Jan, Bing-Yu Chen, Xiang Anthony Chen*, 2026, **CHI'27 Under Review**, Paper.
  #Human-AI Interaction #Animation/VFX #Creative Workflows #Junior-Supervisor Collaboration #AI-Mediated Review

Research Experience

Aalto University, School of Science

*Helsinki, **Finland***

*Research Intern, QuML Group (**1%** acceptance rate; fully funded; applicants from 135 countries)* *Jun. 2026 – Sep. 2026*

- Revised scalable physics-informed neural PDEs/ODEs and generative modeling for complex geometry with Peter Blohm under Prof. Vikas Garg.
- Selected as the first Taiwanese participant in the program's five-year history.

Aalto University, School of Electrical Engineering

*Helsinki, **Finland***

Visiting Researcher, Aalto Acoustics Lab *Jun. 2026 – Sep. 2026*

- Devised supervised flow-matching models for learning room impulse responses for robust sim-to-real transfer and cross-room generalization in room acoustics, collaborated with Kyung Yun Lee, Dr. Nils Meyer-Kahlen, and Dr. Eloi Moliner under Prof. Vesa Välimäki.

RIKEN Center for Computational Science — *Kobe–Osaka, Japan*

*HPC Research Intern (Supercomputer Fugaku), Large-Scale Digital Twin Research Team (**2%** acceptance rate; fully funded)* — *Jul. 2025 – Aug. 2025*

- Researched diffusion neural operators, ODE dynamics, and world models for robust physics-informed sensing under Prof. Hirozumi Yamaguchi.
- Devised graph federated learning for multimodal sensing on supercomputer Fugaku, scaling to 10k clients with enhancing 15x training throughput.

MoonShine Animation Studio — *Taipei, Taiwan*

AI Research Engineer, R& D Team — *Sep. 2025 – Dec. 2026*

- Developed synthetic data-based world model, VLM and 3D/4DGS-based creator tools for animation, VFX, and entertainment production workflows.
- Designed multi-agent MLLM-as-a-judge for efficient artist-supervisor review under Dir. Shaune Jan and Prof. Xiang "Anthony" Chen (UCLA), achieving a 74% preference rate among practitioners.

Delta Electronics — *Taoyuan, Taiwan*

Deep Learning Intern, Power and System Business Group — *Jul. 2023 – Aug. 2023*

- Developed a PyTorch3D point-cloud autoencoder using 74k meshes and an XGBoost surrogate model for finite-element-based magnetic-core design, reducing prediction error to 1% of the formula-based baseline.

Robot-Aided Creation & Construction Research Group — *Tainan, Taiwan*

Research Assistant, Synchronization Lab — *Apr. 2022 – Jun. 2024*

- Developed RL/graph RL for Kuka/UR10/Hiwin manipulators in NVIDIA Isaac Gym/Sim for robotic brick assembly, and ROS2 systems integrating SLAM, YOLO/LSTM, and RL for robot navigation and motion control under Prof. Yang-Ting Shen, Chia-Ching Yen, and Alvin Wen-Yu Su.

Honors & Awards

Foxconn Education Foundation

2026 Foxconn Tech Award — *Jun. 2026*

- Selected for the Foxconn Tech Award ($3,100 USD) for my **undergraduate first-author** publication — combining attention-based *graph clustering* with *reinforcement learning* (Dueling DDQN).

IEEE Signal Processing Society

2023 AIoT Innovation Challenge — *Feb. 2023 – Jun. 2023*

- Secured **3rd Place** & **Sponsors Award** ($3,000 USD) from NXP Semiconductors (Awards).

RIKEN Japan

Overseas Conference Travel Grant — *2026*

- Travel support recipient ($2,200 USD) for attending the ISC High Performance conference 2026 (June 22-26, Hamburg, Germany).

National Science and Technology Council

2025 Overseas Conference Travel Grant — *Dec. 2025*

- Graduate student travel support recipient ($1,000 USD) for SIGGRAPH Asia 2025 conference (December 16 – 18, Hong Kong).

Academic Service

ACM SIGGRAPH Asia 2026 Workshop* | *Lead Organizer — *Dec. 2026*

- Led proposal development, invited-speaker outreach, and program organization for *"Beyond Rigid Bodies × Newton for Robotics: Open GPU Physics, Differentiable Simulation, and Multi-Solver Ecosystems for Physical AI and Embodied Robotics"*, including the integration with NVIDIA's Newton / Physical AI team to connect academic research in differentiable simulation and embodied robotics with GPU-accelerated simulation workflows.
- Organized invited contributions from Taku Komura (HKU), Tao Du (Tsinghua), Srinath Sridhar (Brown), and Yunzhu Li (Columbia), and coordinated **NVIDIA** co-organizers Johnny Nunez Cano, Asier Arranz, and Lu Shi.

IEEE ICRA 2027 Workshop* | *Lead Organizer (Under Review) — *May 2027*

- Led proposal development, technical scope, invited-speaker outreach, and program organization for *"2nd Synthetic Data for Robot Learning: From Synthetic Data to Simulation-Ready Worlds"*, building on the ICRA 2026 SDRL Workshop. Expanded the research scope to simulation-ready 3D generation, differentiable and GPU-accelerated physics, digital twins, and sim-to-real robot learning.
- Coordinated an 11-speaker prospective program spanning academia and industry, including **NVIDIA Physical AI** researchers Johnny Nunez Cano, Asier Arranz, and Lu Shi, alongside Angela Schoellig (TUM), Ming C. Lin (UMD), Beomjoon Kim (KAIST), Jiafei Duan (NUS), and Kento Kawaharazuka (UTokyo). Developed a full-day program featuring invited talks, poster presentations, technical discussions, and simulation-readiness evaluations.

IEEE @ ICRA'27, ITSC'26* | *Reviewer — *Mar. 2026 – Apr. 2026*

- Reviewed **2** full-paper submissions in intelligent transportation systems.

IEEE ICASSP'27** (Audio and Acoustic Signal Processing [AA])* | ***Reviewer — *Oct. 2026 – Present*

- Reviewed **10** full-paper submissions in audio and acoustic signal processing.

ACM @ SIGGRAPH ASIA'26, CHI'25, HRI'25, CSCW'24, WWW'25* | *Reviewer — *Jan. 2024 – Jan. 2025*

- Reviewed **9** full- and short-paper submissions in animation/simulation, human–computer/robot interaction and web retrieval.

***NTU AI Club** @ Taipei, Taiwan | Lead Technical Specialist* *Jan. 2025 – Jun. 2027*

- Spearheaded academic research initiatives and lectures on AI; designed and mentored advanced technical workshops in AI research.

Projects

Generative Modeling on Synthetic Data | *Github: Generative-Modeling-Synthetic-Data* | *Python, PyTorch*

- Implemented and benchmarked *GAN*, *diffusion models* (*DDPM*/*DDIM*), and *Mean Flows* for One-step diffusion on a checkerboard dataset with end-to-end training, sweep-based inference (#steps, η), and comprehensive evaluation using Energy Distance and 2-Wasserstein Distance.
- Conducted ablation studies on timestep embeddings (learned vs. sinusoidal) and analyzed the stochasticity–fidelity trade-off in DDIM via η sweeps; compared sampling efficiency and geometric alignment across methods; made visualizations of distribution dynamic evolution.

Multimodal LLM Agents | *Github: Artificial-Intelligence_Multimodal_Agent* | *Python*

- Built a BLIP+Phi-4 pipeline to caption MSCOCO-Test and Flickr30k with BLEU/ROUGE/METEOR and added Stable Diffusion 3-medium *text-to-image* & SD v1.5 *style transfer*. (Visual 1); Implemented a *RAG* system using Phi-2+all-MiniLM embeddings for resume QA and pytesseract+Phi-4 captioning for PDF indexing. (Visual 2)
- Created a GPT-4o-mini/pyautogen *multi-agent* restaurant rater mapping adjectives to scores and aggregating via geometric mean. (Visual 3)

CAN Bus Scheduling & ADAS Prototype | *Github: Introduction-to-Intelligent-Vehicles* | *Python*

- Applied Simulated Annealing (SA) to optimize priority assignments for CAN bus messages, minimizing total worst-case response-time cost with unschedulability penalties via iterative swaps and Metropolis–Hastings acceptance. (Visual 1)
- Developed KOSOKU, a modular *ADAS* system in *CARLA Simulator* integrating *V2X communication*, multimodal sensor fusion (LiDAR/Radar/Camera), Hybrid A* state-search planning, and OTA map updates for dynamic trajectory generation and replanning(Visual 2).

Multi-Agent RL Communication | *Github: MARL_Communication* | *Python, PyTorch, MuJoCo, Gymnasium*

- Implemented a scalable *MADDPG* baseline for *multi-agent communication* in *MuJoCo locomotion* with a shared actor, central mean-pooled critic, replay buffer, and logging of return, AUC, wall-clock, and memory, including automatic plotting and video for qualitative evaluation.
- Prototyped *Attention/Transformer/GNN-based* multi-agent architectures on top of the MADDPG framework to study coordination and communication, and evaluated generalization across different agent counts and tasks for training and cross-task/variable-N evaluation.

Professional Skills

Coding **Python**, **C/C++**, **C#**, Matlab, LaTeX .
Tools **PyTorch**, **TensorFlow**, OpenCV, OpenGL, WebGL, ROS 1/2, NVIDIA Omniverse Isaac Sim/Gym, Pybullet, CARLA.
Languages Mandarin (native), English (proficient).